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Part A - Visualisation Analysis

1. *Happy People, Happy Planet* (Li & Mendoza, 2023)

Encoding

The visualisation uses several coordinated views. The primary graphic is a world map where each country is represented by an emoji marker. Position encodes geographic location, colour encodes ecological footprint on a three-tier scale (green = low, blue = medium, red = high), emoji expression encodes happiness (smiling to sceptical), and marker size represents population. A scatterplot (upper right) places ecological footprint on the x-axis and happiness on the y-axis, with a regression line supporting correlation reading. Marker shape differentiates developed (triangles) vs. developing (circles). The lower section contains four faceted scatterplots (2011, 2014, 2017, 2020) with the same axes but a simplified colour scheme: green for developing and black for developed; a reference line shows the mean footprint in each year. A partitioned panel separates the two development groups by year, removing country-level detail for a cleaner category comparison.

Data

The dataset is country-level, multivariate and mostly quantitative. Happiness and ecological footprint are continuous numerical variables; population is encoded as marker size; and development status is a categorical attribute. Time is represented by four separate reference years. As a result, the visualisation allows for both cross-sectional (countries at a single point in time) and longitudinal (year-by-year changes) comparisons.

Tasks

The visualisation supports several key analytical tasks:

- Explore geographic patterns of happiness and ecological impact.
- Compare variables (happiness vs. footprint) and recognise correlations or clusters.
- Identify outliers, such as highly happy, low-footprint countries.
- Compare developed vs. developing nations across multiple years.
- Track temporal shifts to see whether happiness or footprint changes over time.

Analysis

The encodings are appropriate for the intended tasks. The map answers "where" questions while also highlighting four variables per country without using excessive text or numbers. Emoji facial expressions and colour make the visualisation intuitive and approachable, allowing for quick identification of broad patterns (such as high-population countries with high ecological footprints). The scatterplots shift the focus from geography to correlation, with position functioning as the most effective channel for quantitative comparison. Faceting prevents overplotting and allows temporal change to be read discreetly. The categorical separation in the lower-right panel clarifies group-level differences by removing individual country noise.

One drawback is the inconsistent colour meaning between panels containing scatterplots, which adds cognitive load to the reader. However, legends and shape encodings help to partially offset this. Although not precise for numerical extraction, the visualisation succeeds as an exploratory multi-view system, communicating patterns in happiness, footprint, population and development status without overwhelming the reader.

2. Sample Superstore - Sales Performance (Pradeep Kumar, 2024)

Encoding

The dashboard combines spatial, temporal and categorical views. The map uses position to locate states and colour on a grey-to-blue scale to encode total sales, where darker blue indicates higher values. Marker size highlights sales volume: all markers are the same colour (semi-transparent white with a blue outline), so colour does not encode any variable. Shown beneath the map is a summary of the four top-selling states' total sales and profit percentages with triangular graphics (green upwards for profit, red downwards for loss).

The Sales by Region panel on the right displays four faceted area charts, one for each region (Central, East, South, and West), with January sales from 2019 to 2023. Sales are encoded by their position on the y-axis, and a profit percentage indicator is denoted beneath each region's name and accompanying graphic.

The Sales Trend by Category panel features three faceted vertical bar charts (Technology, Furniture, and Office Supplies), each representing yearly sales over the same period. Faceting avoids overlapping marks and simplifies category comparison.

The Sales by Sub-Category panel combines two idioms: a heatmap-style plot on the left, where dot intensity reflects sales frequency across years, and a horizontal bar chart on the right, where bar length encodes total revenue per sub-category. Position and length allow for precise magnitude comparison.

Data

The dataset contains U.S. retail performance data, both quantitative (sales and profit) and categorical (state, region, category, and sub-category). Time is represented discretely from 2019 to 2023. The dashboard's multiple panels allow the user to navigate from national-level data to product-level detail, examining patterns across geography, time, and product categorisation.

Tasks

Tasks that can be carried out through the visualisation include:

- Identifying high and low performing states.
- Comparing regions and categories over time to detect trends.
- Separating revenue versus frequency at the product level.
- Detecting profit gains and losses alongside sales totals.
- Exploring whether certain markets consistently outperform others.

Analysis

The encodings are effective for these tasks. The map answers spatial comparison questions and provides an appropriate overview without using numerical labels. Marker size and state colour combine to direct attention to high-revenue states. Faceting is a reliable means of avoiding clutter: regional area charts allow temporal trend comparison, while category-level bar charts provide clear vertical and horizontal magnitude readings. The combined sub-category view separates frequency (heatmap dots) from revenue totals (bar length), allowing for more nuanced comparisons than what a single chart could provide. A slight drawback is the dominance of blue across multiple panels, which may reduce distinction between panels; however, consistent legends and spatial separation minimise confusion. Ultimately, the dashboard uses position, size and faceting to maintain readability while revealing both macro- and micro-level sales patterns across the United States.

3. Musk Tweets (VisualCapitalist, 2022)

Encoding

The visualisation is a radial sunburst chart summarising over 15,000 tweets from 2012 to 2022. At the centre is an image of Elon Musk, surrounded by three primary sectors representing his main discussion areas: SpaceX, Tesla and Other Projects. Colour distinguishes these categories (orange, green and blue, respectively), while radial position encodes hierarchy: inner rings represent broad topics, and outer rings contain more specific sub-topics. Segment size (angular extent and radial length) encodes cumulative keyword frequency, so larger shapes indicate topics mentioned more often across the decade. Time is introduced at the outermost ring, where angular position corresponds to years 2012–2022. Additional annotations, icons and text callouts highlight notable events or spikes in activity. A small “tweets per day” line chart arcs around the outside and adds a second temporal encoding.

Data

The underlying dataset is textual social media data, aggregated from more than 15,000 tweets. Following keyword extraction, tweets are classified into categorical topics (e.g., SpaceX, Tesla, cryptocurrency, manufacturing), with quantitative counts of keyword frequency reflecting how frequently each subject appears. Time is discrete and longitudinal, spanning an entire decade.

Tasks

This visualisation supports the following tasks:

- Identifying which subjects dominate Musk’s public communication.
- Comparing relative importance of topics such as Tesla vs SpaceX.
- Exploring detailed sub-topics within each major category.
- Noticing changes in focus across time, shown by the outer timeline.
- Locating key events using annotation callouts.

Analysis

The encodings used are generally appropriate for an infographic-style summary such as this. Colour clearly distinguishes the three main subject domains, and the radial hierarchy demonstrates part-to-whole structure. Representing time as the outer ring avoids the need for a separate chart and allows readers to trace when subjects gained or lost prominence. The added “tweets per day” line chart compensates for radial imprecision and provides a more accurate trend reading than angle alone. However, radial layouts present comparison challenges: sectors positioned far apart are difficult to compare accurately, and small text in the “Other Projects” branches clutters the visual. Without a numerical scale, readers estimate magnitudes based on area rather than reading exact counts. Time encoded in angles is also less intuitive than a standard x-axis. Despite these drawbacks, the visualisation succeeds in condensing a decade of unstructured text into a compact hierarchical form. It demonstrates narrative insight and thematic structure, revealing how Musk’s public communication shifted over time and which domains received the most attention.

Part B - Visualisation Design**Data: [SteamDB - “Popular Game Franchises on Steam” (*)]**

This dataset lists every Steam game franchise that has collectively accumulated at least 15,000 user reviews across all their games and applications. Each row represents one franchise and provides the following attributes: the franchise’s rank (ordered by total positive reviews), its name, the number of products in the franchise, the total number of positive reviews, the total number of negative reviews, and an overall rating expressed as a percentage. All variables in the dataset are straightforward: the quantitative values are counts of games and review totals, while the categorical variable is franchise name. Because the dataset aggregates multiple games into one franchise entry, it is a high-level summary table rather than a game-level dataset, and it does not include time or sales information.

Tasks

Some visualisation tasks that could be supported by this dataset include:

- Comparing and ranking which franchises are the most positively reviewed, have the highest volume of total feedback, and achieve strong ratings with relatively few products.
- Identifying franchises that are critically well-received versus those that are widely played but polarising.
- Finding correlation trends such as whether a larger franchise (more products) tends to have higher total positive reviews or whether smaller franchises occasionally outperform larger ones?
- Outlier detection, such as a franchise with a high number of products but weak ratings, or a franchise with only a few games but exceptionally high positivity.

Encodings & Analysis

The purpose of the visualisation would be exploratory, allowing users to move beyond a static sorted table and discover patterns between franchise size and community reception. The main visual idiom could be a scatterplot, placing total positive reviews on one axis and total negative reviews on the other. This would show which franchises simply have large communities (high values in both) versus those that are polarising (high negatives relative to positives). Each point would represent a franchise, with marker size encoding the number of products and colour encoding the rating percentage (green for high rating, red for low). This idiom would combine three quantitative attributes (positives, negatives, and catalogue size) and one derived measure (rating) without overwhelming the viewer.

To support the ranking more clearly, a horizontal bar chart could list franchises by total positive reviews. Bars are more effective for accurate value decoding and would make the popularity ordering more interpretable than the table alone. A second bar chart could list franchises by percentage rating, revealing critically strong but smaller franchises. A review ratio plot (positive vs. total) could also be added to show which franchises have overly negative feedback despite their high player base.

These encodings use position for the most important quantitative comparisons, size for supplementary context, and colour for quality differentiation. With simple filtering tools (e.g., search by franchise, hide outliers, show only franchises above a certain rating), this visualisation would reveal relationships that are not visible in a sorted table, especially around quality versus popularity trade-offs.

(*) <https://steamdb.info/franchises/>